

Workplace Hazardous Materials Information System (WHMIS)

WHMIS: Hazard Classification

Certain workers are at risk of hazards from exposure to chemicals. Canada's regulations for hazardous materials, the Workplace Hazardous Materials Information System, or WHMIS, partners with the Globally Harmonized System of Classification and Labelling of Chemicals, or GHS, to protect workers at risk. You need to be aware of the different hazard classifications and how to protect yourself against them so you can stay as safe as possible. This lesson discusses the WHMIS 2015 supplier requirements under federal legislation—the Hazardous Products Act and the Hazardous Products Regulations, or HPR.

Health Canada is the government body responsible for most WHMIS supplier-related laws. Note that WHMIS is also regulated in the workplace by the provincial, territorial, and federal (for federally regulated workplaces) governments through their occupational health and safety legislation. While these jurisdictions mostly based their WHMIS regulations on a common model, small variations between jurisdictions may exist regarding WHMIS 2015.

Suppliers and employers must use and follow the WHMIS 2015 requirements for labels and safety data sheets, also called SDSs, for hazardous products sold, distributed, or imported in Canada.

So, let's get right into it and start with: What is a hazard group?

WHMIS 2015 applies to two major groups of hazards: physical and health. Each hazard group includes hazard classes that have specific hazardous properties.

Physical hazards group: based on the physical or chemical properties of the product—such as flammability, reactivity, or corrosivity to metals.

Health hazards group: based on the ability of the product to cause a health effect—such as eye or respiratory irritation, or if it may cause cancer.

GHS also defines an environmental hazards group. This group, and its classes, were not adopted in WHMIS 2015. However, you may see the environmental classes listed on certain labels and SDSs, although not mandatory, including information about environmental hazards is allowed by WHMIS 2015.

What is a hazard class?

Hazard classes are a way of grouping together products that have similar properties. Most hazard classes are common to GHS and used worldwide by all countries that have adopted GHS. However, some hazard classes are specific to WHMIS 2015.

Common types of hazard classes include flammable gases, flammable aerosols, oxidizing gases, and gases under pressure.

Moving on to hazard categories. Each hazard class contains at least one category, or type. Each category or type is assigned a number and an alphabetical letter.

Some hazard classes contain only one category, such as it causes cancer, whereas other classes contain more than one. The category tells you about how hazardous the product is—that is, the severity of the hazard. Category 1 always represents the highest level of hazard. In cases where Category 1 is further divided, Category 1A within the same hazard class is considered a greater hazard than category 1B. Category 2 is considered more hazardous than Category 3, and so on.

As with all legislation, there are a few exceptions to this rule. For example , for the “Gases Under Pressure” hazard class, the hazard categories are "Compressed Gas", "Liquefied Gas", "Refrigerated Liquefied Gas," and "Dissolved Gas”. These classes do not describe the degree of hazard, but instead relate to the physical state of the gas when packaged.

What are the main concerns for each physical hazard class?

For safety purposes, you should be aware of the different hazard classes and the specific risks related to each one. Learn which hazards your workplace is dealing with and the basic properties of those chemicals so you can react appropriately in case of an accident.

Let's go through some of them.

Flammable classes, are known for obviously presenting the hazards of catching fire or exploding.

Oxidizing gases, liquids, or solids, are oxidizers susceptible of causing or intensifying a fire or causing a fire or explosion.

Gases under pressure include compressed gases, liquefied gases, dissolved gases, and refrigerated liquefied gases. Because of the high pressure inside the cylinder or container, it could explode if heated. Refrigerated liquefied gases are very cold and can cause severe cold, cryogenic, burns, or injury.

- **Self-reactive substances and mixtures include any products that may react on their own to cause a fire or explosion, or that may cause a fire or explosion if heated.**
- **Pyrophoric liquids, solids, and gases are products that can catch fire very quickly and spontaneously if exposed to air.**
- **Self-heating substances and mixtures are products that may catch fire if exposed to air. These products differ from pyrophoric liquids or solids in that they will ignite only after a longer period of time, or when in large amounts.**
- **Substances and mixtures which are in contact with water, emit flammable gases. In some cases, the flammable gases may ignite very quickly and spontaneously.**
- **Organic peroxides may cause a fire or explosion if heated.**
- **Products corrosive to metals. These products may chemically damage or destroy metals.**
- **Combustible dust. This class is used to warn of products that are finely divided into solid particles. If dispersed in air, the particles may catch fire or explode if ignited.**
- **Simple asphyxiants are gases that may displace oxygen in air and cause rapid suffocation.**

- **Physical hazards, not otherwise classified.** This class is a catch-all for hazardous products that don't fit in other categories. To be included in this class, the hazard in question must have the characteristic of occurring by chemical reaction and result in the serious injury or death of a person at the time the reaction occurs. If a product is in this class, the hazard statement on the label and SDS will describe the nature of the hazard.

Now, let's go over the main concerns for each health hazard class.

- **Acute toxicity.** Products in this class are fatal, toxic, or harmful if inhaled, following skin contact, or if swallowed. Acute toxicity can result from exposure to the product itself, or to a product that, upon contact with water, releases a gaseous substance that can cause acute toxicity.
- **Skin corrosion and irritation.** This class covers products that cause severe skin burns and damage.
- **Serious eye damage or eye irritation.** This class covers products that cause serious eye damage and corrosion.
- **Respiratory or skin sensitization.** A respiratory sensitizer is a product that may cause allergy or asthma symptoms or breathing difficulties if inhaled. Skin sensitizer is a product that may cause an allergic skin reaction.
- **Germ cell mutations.** This hazard class includes products that may cause or are suspected of causing genetic defects to body cells that can be passed on to future generations.
- **Carcinogenicity.** This hazard class includes products that may cause or are suspected of causing cancer.
- **Reproductive toxicity.** This hazard class includes products that may damage or are suspected of damaging fertility or the unborn child. Note: there is an additional category which includes products that may cause harm to breast-fed children.
- **Specific target organ toxicity—single exposure.** This hazard class covers products that cause or may cause damage to organs following a single exposure. This class also includes a category for products that cause respiratory irritation, drowsiness, or dizziness.

- **Specific target organ toxicity—repeated exposure.** This hazard class covers products that cause or may cause damage when workers are exposed frequently or for long periods of time.
- **Aspiration hazard.** This class includes products that cause aspiration when a substance, other than air, enters the lungs. Serious health effects can occur such as chemical pneumonia, injury to the lungs, and death.
- **Biohazardous infectious materials.** These materials are microorganisms, nucleic acids, or proteins that either cause or are a probable cause of infection, with or without toxicity, in humans or animals.
- **Physical hazards not otherwise classified.** This class covers products that are not included in any other health hazard class. These hazards have the characteristic of occurring following acute or repeated exposure and have an adverse effect on the health of the person exposed—including injuries or even death. If a product is in this class, the hazard statement will describe the nature of the hazard .

How will you know which hazard classes or categories are assigned to a hazardous product?

The HPR requires suppliers to evaluate products covered by the Hazardous Products Act against specific criteria. If the product meets any of the criteria for a hazard class, it is known as a hazardous product. All hazardous products must be labelled according to applicable regulations and have a corresponding SDS.

The hazard classes and categories will be set out in Section 2 of the SDS, the Hazard Identification section. Each hazard class or category must use specific pictograms and other label elements to indicate the hazard that is present, and which precautionary measures should be taken. It's important to use the information provided by the label and SDS in order to be informed and to know how to safely use, handle, store and dispose of hazardous products.

So, now that you know some of the different classes and categories, as well as information on how to stay safe, how can you apply this knowledge in your workplace? What do you need to do to protect yourself from exposure, and how can you best react if an emergency occurs? Keep that in mind when doing your everyday activities in your role.